Mobile Computing and Its Applications
Mini-Project Proposal

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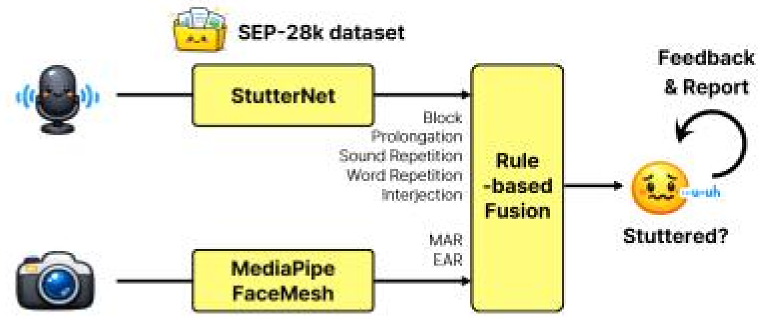

■ Goal & Usage Scenario

Stuttering is a speech fluency disorder affecting approximately 1% of the global population, characterized by involuntary disruptions such as sound repetitions, prolongations, and blocks. It is a condition specific to people with speech disorders, but it is also a general concern for anyone speaking under stress or anxiety—such as non-native English speakers preparing for presentations. Real-time awareness of disfluency patterns is critical in both context, yet rarely available outside of clinical settings.

FluentSpeech is an Android app that provides real-time stuttering detection and feedback during speech. The user activates the app in any desired speaking situation—whether as part of daily monitoring for people who stutter, or during English presentation practice for general users. The app simultaneously captures audio to detect disfluency events via an optimized StutterNet (TDNN-based model trained on SEP-28k), while the front camera with MediaPipe FaceMesh extracts facial tension cues as an auxiliary signal. When stuttering is detected, a feedback message appears on screen; when disfluency decreases, positive feedback is displayed. As the session ends, a summary report shows the frequency of disfluency.

- **Usage Scenario**: A user preparing for an English presentation launches FluentSpeech and begins speaking while holding the phone naturally. The app processes audio in 3-second sliding windows, detecting disfluency events in real time. Color-coded on-screen messages appear when stuttering is detected, and fade when fluency recovers. A facial tension gauge derived from lip and eye landmarks serves as a secondary indicator of speech anxiety. At session end, a report displays the count of each disfluency detected, allowing users to review their own speech patterns.

■ Related Works

- **SEP-28k (Lea et al. (2021))** : Using 28,177 three-second clips from 385 podcasts labeled across five disfluency types, they trained LSTM and ConvLSTM models on mel-spectrogram and pitch features to classify Block, Prolongation, Sound Repetition, Word Repetition, and Interjection.

- **FluentNet (Kourkounakis et al. (2021))** : Trained on the UCLASS dataset and the synthetic LibriStutter corpus, the model takes STFT(Short-Time Fourier Transform) spectrograms as input and passes them through SE-ResNet blocks, BiLSTM layers, and a global attention mechanism to classify six disfluency types: Sound/Word/Phrase Repetition, Revision, Interjection, and Prolongation.

- **StutterNet (Sheikh et al. (2021))** : Using MFCC features extracted from the UCLASS dataset, five TDNN layers capture temporal context across multiple time windows, followed by statistical pooling and fully connected layers to classify all disfluency types jointly in a single lightweight model.

- **MMSD-Net (Nie et al. (2024))** : Combining SEP-28k, FluencyBank, Adults Who Stutter, and SpeakingFaces datasets, the model fuses audio, video, and text through independent Transformer encoders and a Q-Former fusion module. The paper demonstrates that facial expression signals carry meaningful information for stuttering detection.

- **Cross-lingual wav2vec2 Transfer (Grosman et al. (2025))** : Fine-tuning 15 wav2vec2-based models on raw audio from 18 languages drawn from Common Voice, the study evaluates cross-lingual transfer via character error rate. The key finding was that non-Indo-European languages (like Korean) benefit significantly less from English-trained models.

■ Key Idea

FluentSpeech uses a dual-modal pipeline combining an audio branch for disfluency classification and a visual branch for facial tension estimation.

- **Audio Branch (Microphone)** : 20-dimensional MFCC features from a 3-second sliding window are fed into a TDNN-based model trained on SEP-28k dataset, classifying five disfluency types: Block, Prolongation, Sound Repetition, Word Repetition, and Interjection. The model undergoes structured channel pruning, and is converted to TFLite with INT8 quantization.

- **Visual Branch (Camera)** : MediaPipe FaceMesh extracts 468 facial landmarks in real time. Mouth Aspect Ratio(MAR) and Eye Aspect Ratio(EAR) are computed rule-based and normalized into a tension score.

- **Feedback & Interaction** : When a disfluency is detected, a negative warning appears on the screen. When fluency recovers, positive feedback is displayed. As a possible extension (if time permits), a touch or shake interaction will be added. The user can tap the screen or shake the phone to trigger a pacing aid (rhythm guide or breathing cue) that helps them regain their speaking flow, leveraging the touchscreen or accelerometer sensor.

■ Implementation Plan

Mobile App Development	- Android Studio (Kotlin)
Dataset	- **SEP-28k** https://github.com/apple/ml-stuttering-events-dataset/
Model	- **StutterNet** + Optimization (Pruning/Quantization) https://arxiv.org/abs/2105.05599
Library	- MediaPipe (Android SDK) - librosa (MFCC Feature)
Mobile Sensors (API)	- AudioRecord API - CameraX API

[ML]

1) Extract MFCC features from SEP-28k dataset
2) Train TDNN model (StutterNet)
3) Apply channel pruning & INT8 quantiztion, Convert to TFLite
4) Implement Android audio pipeline
5) Integrate CameraX and MediaPipe, Compute MAR/EAR tension score
6) Implement Rule-based Fusion

[UI]

1) Implement basic UI using microphone and camera
2) Implement feedback UI (visualizing sensor data)
3) Implement report UI
4) Implement interaction UI (if time permits)

[Other works]

1) Testing models and application
2) Demo recording
3) Preparing final presentation

■ Project Timeline

	Date	Task
1	5/5 ~ 5/11	**[ML-1, 2]** Traning Models **[UI-1]** Basic UI Implementation
2	5/12 ~ 5/18	**[ML-3, 4]** Optimization and Audio Pipeline **[ML-5]** Camera Pipeline
3	5/19 ~ 5/25	**[ML-6]** Rule-based Fusion **[UI-2, 3]** Feedback and Report UI Implementation
4	5/26 ~ 6/1	**[UI-4]** Interaction UI *(if time permits)* **[Other Works]** Testing and Demo Recording

■ Github Link : https://github.com/pcy1203/FluentSpeech

Reference

Lea, C., Mitra, V., Joshi, A., Kajarekar, S., & Bigham, J. P. (2021, June). **Sep-28k: A dataset for stuttering event detection from podcasts with people who stutter**. In ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 6798-6802). IEEE.

Kourkounakis, T., Hajavi, A., & Etemad, A. (2020). **FluentNet: end-to-end detection of speech disfluency with deep learning**. arXiv preprint arXiv:2009.11394.

Sheikh, S. A., Sahidullah, M., Hirsch, F., & Ouni, S. (2021, August). **Stutternet: Stuttering detection using time delay neural network**. In 2021 29th European signal processing conference (EUSIPCO) (pp. 426-430). IEEE.

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Grosman, J., Almeida, C., Schardong, G., & Lopes, H. (2025). **On the Cross-lingual Transferability of Pre-trained wav2vec2-based Models**. arXiv preprint arXiv:2511.21704.